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This TOP supersedes TOP 02-2-602, Acceleration; Maximum and Minimum Speeds, dated 28 January 1981.

Marginal notations are not used in this revision to identify changes, with respect to the previous issue, due to the extent of the changes.

14. ABSTRACT
This TOP prescribes the procedures used for determining the acceleration profile and maximum and minimum speeds of wheeled and tracked vehicles.

15. SUBJECT TERMS
acceleration, minimum speed, maximum speed, tracked vehicle, wheeled vehicle, throttle, automatic transmission, gear shift

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U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

*Test Operations Procedure 02-2-602A

25 March 2019

DTIC AD No. AD1070440

ACCELERATION; MAXIMUM AND MINIMUM SPEEDS

			Page
Paragraph	1.	SCOPE	2
	2.	FACILITIES AND INSTRUMENTATION	2
	2.1	Facilities	2
	2.2	Calibration	2
	2.3	Instrumentation	3
	2.4	Data Acquisition	3
	3.	REQUIRED TEST CONDITIONS	4
	3.1	Preparation for Test	4
	3.2	Test Controls	4
	3.3	Test Restrictions	5
	4.	TEST PROCEDURES	5
	4.1	Maximum Speed Acceleration Test	5
	4.2	Minimum Speed Test	6
	5.	DATA REQUIRED	7
	5.1	Maximum Speed Acceleration Test	7
	5.2	Minimum Speed Test	7
	6.	PRESENTATION OF DATA	7
	6.1	Maximum Speed Acceleration Test	7
	6.2	Minimum Speed Test	9
APPENDIX	A.	ABBREVIATIONS	A-1
	B.	REFERENCES	B-1
	C.	APPROVAL AUTHORITY	C-1

*This TOP supersedes TOP 2-2-602, Acceleration; Maximum and Minimum Speeds, dated 28 January 1981.

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1. SCOPE.

This Test Operations Procedure (TOP) describes methods used to measure, analyze, and report the acceleration profile, maximum speed, and minimum speed of wheeled and tracked vehicles.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

Item	Requirement
Level, Paved Road (i.e., U.S. Army Aberdeen Test Center (ATC) Perryman 3-mile Straightaway, ATC Phillips Army Airfield (PAAF) Main Runway, ATC Automotive Technology Evaluation Facility (ATEF) paved track straight section, Yuma Test Center (YTC) Laguna Paved course, YTC Laguna High Speed Paved Oval, YTC General Motors (GM) Performance Straight Track)	A straight, level, paved road with a lane width of not less than 3.7 meters (m) (12.0 feet (ft)), a longitudinal gradient $\leq$ 1 percent, and a super elevation of $\leq$ 2 percent.

2.2 Calibration.

a. All measuring tools and instrumentation will be calibrated against a higher order standard at periodic intervals not to exceed twelve months. Records showing the calibration traceability to the National Institute of Standards and Technology (NIST) will be maintained for all measuring and test equipment.

b. All measuring and test equipment will be labeled with the following information:

(1) Date of calibration.

(2) Date of next scheduled calibration.

(3) Name of the organization and technician who calibrated the equipment.

c. A written calibration report will be provided that includes as a minimum the following information for all measurement and test equipment:

(1) Type of equipment, manufacturer, model number, etc.

(2) Measurement range.

(3) Accuracy.

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(4) Calibration interval.

(5) Type of standard used to calibrate the equipment (calibration traceability of the standard must be evident).

2.3 Instrumentation.

Devices for Measuring	Permissible Measurement Uncertainty
Road speed (velocity)	1%
Distance	1%
Engine speed	1%
Time	1%
Meteorological data:	
Ambient temperature	1 °Celsius (°C)
Wind speed	5%
Wind direction	50 milliradian (mrad)

NOTE: The permissible measurement uncertainty is the two-standard deviation value for normally distributed instrumentation calibration data. Thus 95% of all instrumentation calibration data readings will fall within two standard deviations from the known calibration value.

a. If the test vehicle is equipped with a Controller Area Network (CAN) data bus, applicable data should be monitored and recorded, as described in TOP 01-2-506[1**]. The data bus can provide key information for acceleration tests, including engine speed, gear ranges selected and attained, torque converter status, throttle position, relative power and torque values, traction control status, etc. If CAN data are used for a key measurement (e.g., engine speed), the validity of the data bus information must be verified and the source of data documented.

b. A global positioning systems (GPS) that uses the Doppler shift effect to measure vehicle road speed may be used provided they meet the measurement uncertainty requirement above. A GPS that uses vehicle position differentiation to estimate road speed may also be used if the sample rate is greater than or equal to 20 Hertz (Hz) and they meet the measurement uncertainty requirement.

2.4 Data Acquisition.

A digital data acquisition system shall be used to record pertinent vehicle data throughout testing, including road speed, engine speed, and CAN bus signals. The sample rate shall be at least 20 Hz.

** Superscript numbers correspond to Appendix B, References.

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3. REQUIRED TEST CONDITIONS.

3.1 Preparation for Test.

a. Review the detailed test plan (DTP) and all instructional material issued with the test vehicle by the manufacturer, contractor, or government, as well as reports of previous similar tests on the same types of vehicles.

b. Select the applicable test procedures to be used based on the requirement documents and purpose of the test.

c. Prepare data collection sheets to record all pre-test information, conditions of test, test results, observations, and measurements that would be valuable for analysis.

d. Ensure that all test personnel are familiar with the required technical and operational characteristics of the test item, test equipment, and test procedures.

e. Ensure all safety Standard Operating Procedures (SOPs) have been reviewed and will be observed throughout test operations.

f. Ensure the test instrumentation and data acquisition systems are functioning properly and calibrated.

g. Select test course and establish start of run and end of run locations.

3.2 Test Controls.

Prior to test initiation, ensure that:

a. The vehicle has been prepared and equipped in accordance with standard use and/or within the specifications presented in the test plan.

b. The vehicle is payloaded in accordance with the test plan.

c. Vehicle fluid levels, track tension, tire pressures, and/or suspension ride height are properly set in accordance with vehicle specifications and the test plan. If the vehicle is equipped with a Central Tire Inflation System (CTIS), verify that the correct setting has been selected for operation on paved roads, and based on the weight configuration of the test item.

d. Tires/track pads, brake, steering, and suspension components are in good serviceable condition. If the vehicle is equipped with an active suspension, verify that the correct setting has been selected for operation on paved roads and in accordance with the test plan.

e. All controllable engine accessory loads (e.g., engine cooling fan, power take-off (PTO), auxiliary generator, etc.), stability systems (e.g., electronic stability control (ESC), traction control, etc.), and electrical accessories (e.g., air conditioning compressor, fans,

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interior/exterior lights, driver's/commander's displays, communication equipment, etc.) are in their operating states specified in the test plan.

f. Vehicle drivetrain, suspension, steering, tire/track, and other applicable components are warmed up to normal operating temperature by driving the test item in a normal operating mode for a minimum of 30 minutes. Alternative schedules that provide equivalent vehicle warm-up can be substituted.

g. A reliable method of communication is established between the test engineer/technician and the vehicle operator, preferably via two-way radio.

3.3 Test Restrictions.

a. Tests are not conducted at night, during inclement weather, or when the road surface may introduce a hazard to the test vehicle or other traffic on the road. Dry, unobstructed surfaces are used unless the test plan introduces a specific requirement. Local safety and operational procedures will be carefully followed. Recommended environmental conditions for test conduct are as follows:

(1) Wind speed: $\leq$ 6.7 meters per second (m/s) (15 mph) average value and $\leq$ 8.9 m/s (20 mph) gust.

(2) Ambient temperature greater than 0 $^{\circ}$C (32 °Fahrenheit (°F)).

(3) Humidity: $\leq$ 95%.

Note: It is also recommended that wind speed in the direction of vehicle travel be monitored and reported with individual test trial results.

b. If military personnel are required for testing, determine if Military Occupational Specialty (MOS) qualified Soldier-Operator/-Maintainer Test and Evaluation (SOMTE) personnel assigned to the U.S. Army Test and Evaluation Command (ATEC) are available to support the testing. If SOMTE are not available, ensure a Test Schedule and Review Committee (TSARC) request is submitted one year prior to the start of testing, or as early as possible. A Safety Release (SR) and Human Research Protection Plan (HRPP) must be obtained from ATEC prior to using military personnel as test participants.

4. TEST PROCEDURES.

4.1 Maximum Speed Acceleration Test.

a. From a standing start, with the engine idling, accelerate the vehicle to the maximum road speed in each gear/range combination required by the test plan by applying wide open throttle (WOT). The vehicle operator should apply the minimum steering input necessary to maintain the vehicle in straight line operation. Ensure that the distance traveled for each test run is sufficient for the vehicle to reach and maintain its maximum road speed in each gear/range

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combination. For the purposes of this test standard, the term throttle refers to the means of controlling or regulating vehicle power, such as throttle, accelerator, pedal, etc. WOT refers to the position of the controlling mechanism which allows maximum power output of the vehicle.

b. Repeat the acceleration test in the opposite direction, with a heading 180 degrees from the direction of the initial test run to negate wind conditions.

c. Conduct a minimum of three test runs in each direction. Conduct the test over the same section of the test course, using the same start point for each run in the same direction.

d. If required by the test plan, repeat the above procedure in the reverse direction.

e. If required by the test plan or requested by the test sponsor, conduct the above procedure from a "brake stand" start. Brake stand starts are conducted by holding the vehicle stationary with the service brakes while applying throttle to achieve a predetermined engine speed. Once that engine speed is achieved, rapidly release the service brakes while applying WOT.

f. For a vehicle with a manual transmission, use the above procedures with the following modifications: the operator shall shift in a manner to optimize acceleration performance without exceeding the maximum recommended engine speed in each gear, with the exception of the highest gear. If possible, consult the engine torque curves (if available from the engine manufacturer) to determine the optimal shift points in each gear. If the shift points in any test run fail to optimize acceleration performance in the judgment of the operator and/or field engineer, the run should be repeated.

g. For a vehicle in which propulsion energy is supplied via electric/traction motor (such as electric, hybrid electric, fuel cell electric, etc.), use the above procedure with the following additions: if applicable, measure and monitor the battery/ultra-capacitor state-of-charge (SOC). Conduct each test run at the normal maximum SOC of the vehicle, in accordance with the manufacturer's specifications. Testing may be conducted at alternate SOCs based on the test plan and vehicle requirements.

4.2 Minimum Speed Test.

a. From a standing start, with the engine idling, release the service brakes and allow the vehicle to travel at idle speed in each gear/range combination required by the test plan. For vehicles requiring the operator to apply throttle to overcome exhaust brakes or transmission brakes, the operator should apply the minimum amount of throttle necessary to maintain forward motion. The vehicle operator should apply the minimum steering input necessary to maintain the vehicle in straight line operation. Ensure that the distance traveled for each test run is sufficient for the vehicle to reach and maintain its idle speed in each gear/range combination.

b. Repeat the minimum speed test in the opposite direction, with a heading 180 degrees from the direction of the initial test run to negate wind conditions.

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c. Conduct a minimum of three test runs in each direction. Conduct the test over the same section of the test course, using the same start point for each run in the same direction.

d. If required by the test plan, repeat the above procedure in the reverse direction.

e. For a vehicle with a manual transmission, use the above procedure with the following modification: the minimum road speed in each gear shall be the road speed corresponding with the lowest engine speed that results in smooth operation (no engine knocking or drivetrain bucking) of the vehicle powertrain.

f. For a vehicle in which propulsion energy is supplied via electric/traction motor (such as electric, hybrid electric, fuel cell electric, etc.), use the above procedure with the following additions: if applicable, measure and monitor the battery/ultra-capacitor SOC. Conduct each test run at the normal maximum SOC of the vehicle, in accordance with manufacturer specifications. Testing may be conducted at alternate SOCs based on the test plan and vehicle requirements.

5. DATA REQUIRED.

5.1 Maximum Speed Acceleration Test.

NOTE: Acceleration time shall be measured from the instant vehicle motion begins. If system-response lag is significant, acceleration time should also be measured from the instant the accelerator is moved. The method used shall be documented.

a. Record vehicle road speed and engine speed versus time with each channel sampled at a minimum of 20 Hz.

b. Maximum road speed in each gear/range combination.

c. Maximum engine speed in each gear/range combination.

5.2 Minimum Speed Test.

a. Minimum road speed in each gear/range combination.

b. Minimum engine speed in each gear/range combination.

6. PRESENTATION OF DATA.

6.1 Maximum Speed Acceleration Test.

a. Plot vehicle road speed versus time for each test run, as shown in Figure 1.

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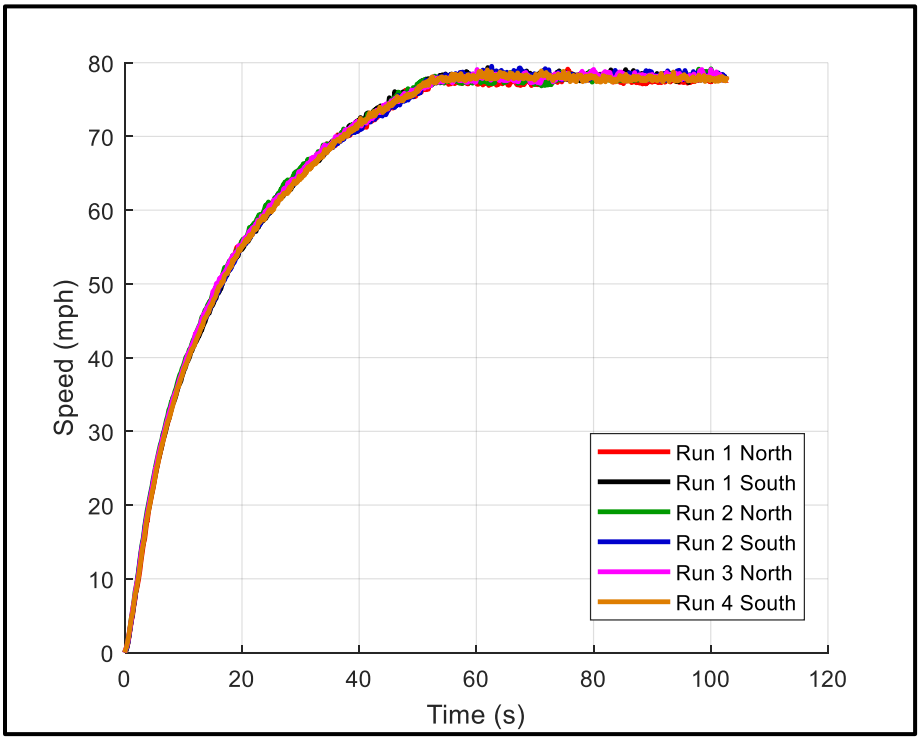


Figure 1. Sample acceleration profile.

b. Present time-to-speed results (in tabular format) in 8 kilometer per hour (km/hr) (5 miles per hour (mph)) increments, unless specified otherwise in the test plan. The time-to-speed is defined as the time (in seconds) from the first vehicle motion until the vehicle first reaches the applicable road speed. The time-to-speed results shall first be determined for each test run using standard data processing methods (e.g., smoothing of time history data). The time-to-speed results for each speed increment shall then be averaged across all the test runs. The final (or maximum) time-to-speed increment presented shall be the highest road speed increment the vehicle achieved during every test run. A sample table with time-to-speed results is presented in Table 1.

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TABLE 1. ACCELERATION TIME-TO-SPEED RESULTS

ROAD SPEED		AVERAGE TIME TO SPEED (seconds)	
km/hr	mph	Configuration 1	Configuration 2
8	5	1.0	1.1
16	10	1.8	1.8
24	15	2.9	2.9
32	20	4.0	4.0
40	25	5.0	5.0
48	30	6.1	6.1
56	35	7.5	7.4
64	40	9.2	9.0
72	45	10.9	10.7
80	50	12.7	12.4
89	55	15.0	14.5
96	60	17.8	17.2
105	65	20.6	20.0
113	70	23.5	22.7

c. The maximum road speed and engine speed presented for each gear/range combination will be the average maximum speed across all the test runs. The maximum road speed is defined as the speed limited either by governor actuation or available vehicle power. The maximum road speed and engine speed for each test run will be determined by averaging the corresponding data channels over a sufficient time interval once the stabilized maximum speeds have been achieved.

6.2 Minimum Speed Test.

Present the minimum road speed and engine speed for each gear/range combination in narrative or tabular format, as appropriate.

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APPENDIX A. ABBREVIATIONS.

ATC	U.S. Army Aberdeen Test Center
ATEC	U.S. Army Test and Evaluation Command
ATEF	Automotive Technology Evaluation Facility
°C	degrees Celsius
CAN	controller area network
CTIS	central tire inflation system
ESC	electronic stability control
°F	degrees Fahrenheit
ft	feet
GM	General Motors
GPS	global positioning system
HRPP	Human Resource Protection Plan
Hz	Hertz
ISO/TR	International Standards Organization Technical Report
km/hr	kilometers per hour
m/s	meters per second
MOS	Military Occupational Specialty
mph	miles per hour
mrad	milliradian
NIST	National Institute of Standards and Technology
PAAF	Phillips Army Airfield
PTO	power take-off
SAE	Society of Automotive Engineers
SOC	state-of-charge
SOMTE	Soldier-Operator/-Maintainer Test and Evaluation
SOP	Standard Operating Procedure
SR	Safety Release
TOP	Test Operations Procedure
TSARC	Test Schedule and Review Committee
WOT	wide open throttle
YTC	Yuma Test Center

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APPENDIX B. REFERENCES.

1. TOP 01-2-506, Use of Controller Area Network (CAN) Data to Support Performance Testing, 16 July 2015.

For information only (related publications).

a. Society of Automotive Engineers (SAE) Standard J1491, Vehicle Acceleration Measurement, July 2006.

b. International Standards Organization Technical Report (ISO/TR) 11954, Fuel cell road vehicles – Maximum speed measurement, October 2008.

c. TOP 01-1-011B, Vehicle Test Facilities at Aberdeen Test Center and Yuma Test Center, 12 December 2017.

d. TOP 02-2-505A, Inspection and Preliminary Operation of Vehicles, 22 October 2018.

e. TOP 02-2-704A, Tires, 15 December 2015.

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APPENDIX C. APPROVAL AUTHORITY.

CSTE-TM 25 March 2019

MEMORANDUM FOR

Commanders, All Test Centers
Technical Directors, All Test Centers
Directors, U.S. Army Evaluation Center
Commander, U.S. Army Operational Test Command

SUBJECT: Test Operations Procedure (TOP) 02-2-602A Acceleration; Maximum and Minimum Speeds, Approved for Publication

1. TOP 02-2-602A Acceleration; Maximum and Minimum Speeds, has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency. The scope of the document is as follows:

This TOP describes methods used to measure, analyze, and report the acceleration profile, maximum speed, and minimum speed of wheeled and tracked vehicles.

2. This document is approved for publication and will be posted to the Reference Library of the ATEC Vision Digital Library System (VDLS). The VDLS website can be accessed at https://vdls.atc.army.mil/.

3. Comments, suggestions, or questions on this document should be addressed to U.S. Army Test and Evaluation Command (CSTE-TM), 6617 Aberdeen Boulevard-Third Floor, Aberdeen Proving Ground, MD 21005-5001; or e-mailed to usarmy.apg.atec.mbx.atec-standards@mail.mil.

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RAYMOND G. FONTAINE
Associate Director, Test Management Directorate (G9)

FOR

MICHAEL J. ZWIEBEL
Director, Test Management Directorate (G9)

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